

Development of an Integrated Safety Monitoring System for Coal Miners using Smart Helmets

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Abstract—The Smart Helmet for Coal Mining through IoT Monitoring project is to improve safety and efficiency in coal mining operations by integrating IoT technology into conventional mining helmets. Coal mining is a dangerous occupation that often takes place in challenging environments, making it crucial to monitor worker safety and environmental conditions. The realization of the smart helmet project, featuring an array of sensors such as DHT11 and MQ135 and IoT functionalities, has ushered in a new era of workplace safety. This innovative device continuously monitors vital parameters like temperature, humidity, and gas levels in real-time inside the coal mine, safeguarding the health and security of workers across various industries. The data collected by these sensors will be wirelessly transmitted to a central monitoring system (CMS) using IoT protocols. The CMS also acts as a centralized dashboard for supervisors and safety personnel to oversee environmental conditions about potential hazards. The platform displays the real-time readings of the temperature, humidity, and gas level in the mine environment also with visualization tools like graph for easy analysis and interpretation.

Keywords — Smart Helmet, IoT Monitoring, Coal Mining Safety, Real-time Data, Sensor Integration, Central Monitoring System.

I. INTRODUCTION

Coal mining is a highly hazardous industry characterized by unpredictable environmental conditions and significant safety risks. Despite advancements in safety protocols and protective equipment, mining operations continue to face challenges in mitigating occupational hazards and preventing incidents. Ensuring the safety and well-being of workers requires innovative approaches that integrate advanced technologies to enhance monitoring and risk assessment. The Smart Helmet for Coal Mining through IoT Monitoring is designed to address these challenges by leveraging the

capabilities of the Internet of Things (IoT) for real-time environmental monitoring and improved decision-making. This system incorporates IoT-enabled sensors, such as the DHT11 for temperature and humidity detection and the MQ135 for air quality assessment, including the detection of hazardous gases like carbon dioxide and ammonia. These sensors continuously collect environmental data, which is wirelessly transmitted to a central server or cloud platform, such as ThingSpeak, using IoT communication protocols. Authorized personnel can securely access real-time data through a web interface, enabling remote monitoring of mine conditions and allowing for timely intervention when hazardous situations arise. An intuitive dashboard displays sensor readings, including temperature, humidity, and oxygen levels, providing a comprehensive overview of workplace safety conditions. In addition to real-time monitoring, it addresses several critical challenges associated with implementing IoT-based solutions in mining environments, including sensor limitations, scalability, robustness, and power efficiency. To improve accuracy and reliability, the system integrates sensor calibration techniques and redundancy mechanisms to reduce errors caused by extreme environmental conditions. The design ensures scalability, enabling deployment across multiple mining sites while maintaining centralized data access and analysis. Given the harsh and dynamic nature of underground mines, the system is built for robustness, incorporating durable materials and protective enclosures to withstand high humidity, dust, and temperature fluctuations. Furthermore, optimized power management techniques, including energy-efficient communication modules and low-power sensors, enhance battery life, reducing the need for frequent maintenance and ensuring continuous operation.

By integrating IoT technology with advanced sensor networks, this Smart Helmet Monitoring System enhances workplace safety, operational efficiency, and risk mitigation in coal mining. The proposed solution not only provides real-time hazard detection and worker protection but also contributes to the broader digital transformation of mining operations.

II. LITERATURE REVIEW

Sandaruwan, Lahiru, and Hettige et al. discuss the design and implementation of smart helmets aimed at improving miner safety [1]. Their study highlights key aspects such as air quality monitoring techniques, proximity detection mechanisms, and localization systems. The research emphasizes the integration of advanced communication technologies, including LoRaWAN, RF communication, and wireless sensor networks, to establish smart alarming zones in mining environments. Although the concept of a smart helmet for coal miners presents a promising solution, further advancements are necessary, particularly in the seamless integration of multiple sensors while ensuring user comfort.

Miners traditionally wear safety helmets, which have been enhanced with technological components to serve as smart protective gear [2]. The mining industry experiences hazardous conditions leading to injuries and fatalities, necessitating the deployment of safety monitoring systems. Smart helmets incorporate gas sensors capable of detecting harmful gases such as carbon monoxide (CO) and methane (CH₄). Additionally, infrared sensors ensure the proper wearing of helmets, while an integrated alarm system—comprising an audible buzzer and LED indicators—activates upon exceeding predefined safety thresholds, providing real-time warnings to miners and their supervisors. Despite the advantages, real-time data management and analysis remain critical challenges for optimizing system performance.

In reference [3], a proposed coal mining safety cap is designed to provide real-time hazard detection and preventive alerts. The helmet continuously records relevant environmental data and updates its software for enhanced monitoring capabilities. Leveraging the Internet of Things (IoT), these wearable devices facilitate seamless data transmission and retrieval from multiple sources. Upon detecting potential risks, alert notifications are dispatched to both miners and mining operators. The advancements in wearable computing and connectivity enable these systems to integrate various sensors for real-time hazard mitigation. However, ensuring data privacy and security remains a key area for future development.

Reference [4] describes a Coal Mining Head Protector designed to enhance miner safety by continuously updating and transmitting real-time data to a cloud platform. IoT-based communication enables seamless data sharing between miners and supervisors, ensuring immediate response to hazardous situations. Alerts are generated and transmitted to the relevant personnel in the event of

potential dangers. While this wearable safety solution significantly improves monitoring capabilities, the high implementation cost remains a critical factor that requires further optimization.

Improving worker safety in mining environments is a primary concern, considering the high-risk conditions, substantial financial investments, and unpredictable operational outcomes associated with mining activities [5]. The proposed smart helmet integrates a worker position detection system, facilitating real-time tracking within underground tunnels and mining lanes. In emergency situations, RF signals and Wi-Fi communication enable the detection and localization of trapped workers. Additionally, gas sensors embedded in the helmet detect the presence of flammable gases (e.g., LPG) and transmit alerts via Wi-Fi to the control station. While the proposed system offers a robust safety framework, the user interface requires further refinement to ensure accessibility and ease of use during critical emergencies.

III. METHODOLOGY

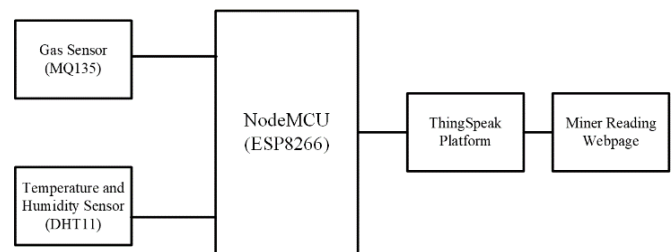


Fig 1: Block diagram of IoT-Based Mine Monitoring System

Fig.1 illustrates the working architecture of a smart miner safety system utilizing IoT-based environmental monitoring. The system integrates two primary sensors: the MQ135 gas sensor and the DHT11 temperature and humidity sensor. The MQ135 sensor is responsible for detecting hazardous gases such as carbon monoxide (CO) and methane (CH₄), which are common in mining environments. Meanwhile, the DHT11 sensor continuously monitors temperature and humidity levels to ensure safe working conditions for miners. Both sensors are interfaced with a NodeMCU (ESP8266), a microcontroller with built-in Wi-Fi capabilities, which serves as the central processing unit for data acquisition and transmission.

NodeMCU collects real-time sensor data and transmits it to the ThingSpeak platform, an IoT analytics service that processes and visualizes data. This cloud-based platform facilitates remote monitoring and analysis, allowing mining supervisors to track environmental conditions in real-time. The data is then displayed on a miner reading webpage, providing users with an intuitive interface for accessing crucial safety information. This system enhances underground safety by enabling proactive hazard detection and timely response mechanisms, ensuring improved working conditions for miners.

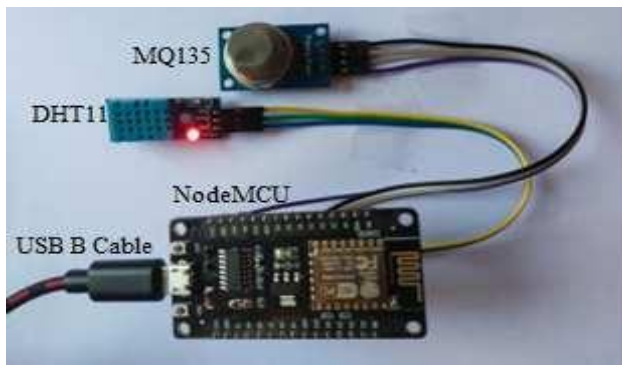


Fig 2: Connection of the hardware components

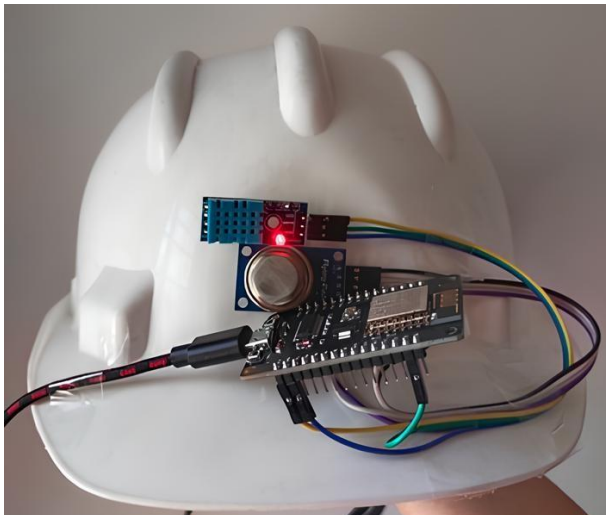
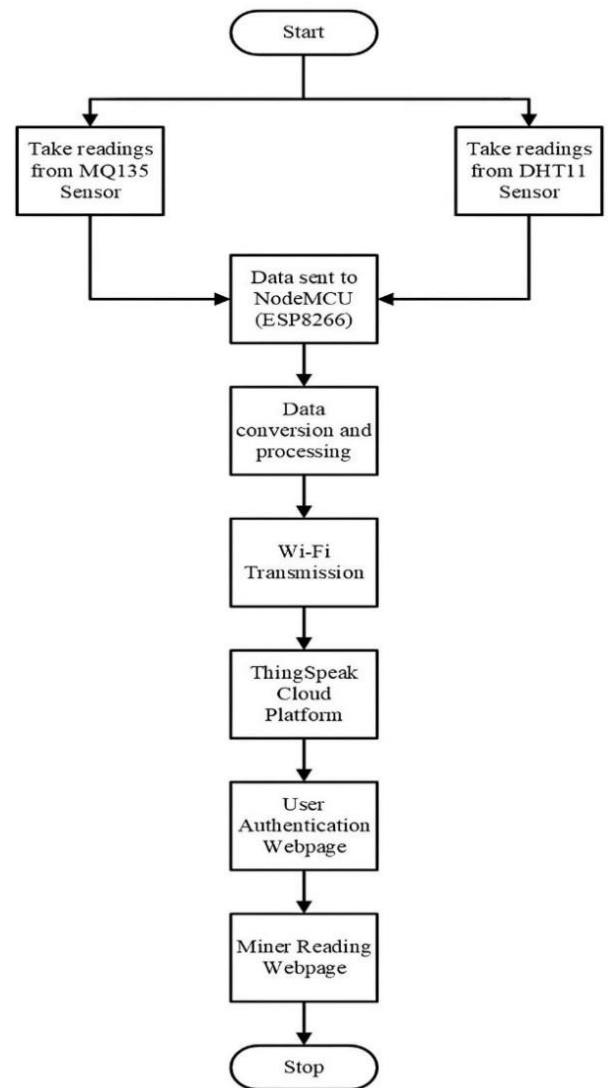


Fig 3: Hardware components integrated on miners' helmet

Fig.2, 3 illustrates the hardware setup of an IoT-based miner safety monitoring system, consisting of a NodeMCU (ESP8266) microcontroller, an MQ135 gas sensor, and a DHT11 temperature and humidity sensor. The connections are established using jumper wires, ensuring seamless communication between the components. The MQ135 gas sensor is connected to the NodeMCU to detect harmful gases such as carbon monoxide (CO) and methane (CH₄), providing crucial air quality data. Similarly, the DHT11 sensor is wired to the NodeMCU to measure temperature and humidity levels, ensuring environmental conditions remain within safe limits.

Power is supplied to the system via a USB B cable connected to the NodeMCU, which also enables data transmission through its built-in Wi-Fi module. The collected sensor readings are transmitted to the ThingSpeak cloud platform for remote monitoring and analysis. This setup forms the core of a smart helmet system, enhancing miner safety through real-time hazard detection and alert mechanisms, with the potential for integration into a comprehensive industrial safety framework.



communication in Smart Helmet Safety System

Fig.4 shows the following sequential steps that are involved - The data is collected from MQ135 and DHT11 sensors, respectively in the environment. The analog signals collected from the sensors are sent to the NodeMCU (ESP8266). This data is then converted into digital format and processed in the NodeMCU microcontroller. This microcontroller then connects to Wi-Fi to transmit the data to the ThingSpeak platform. ThingSpeak stores, reads, and analyses the data. Before displaying the data to users, the system urges user authentication through a login page. The user must enter the correct username and password to access the miner reading page. Once authentication is verified, the users are directed to the Miner Reading Webpage, where they can access the Realtime data of the environment.

IV. RESULT AND DISCUSSION

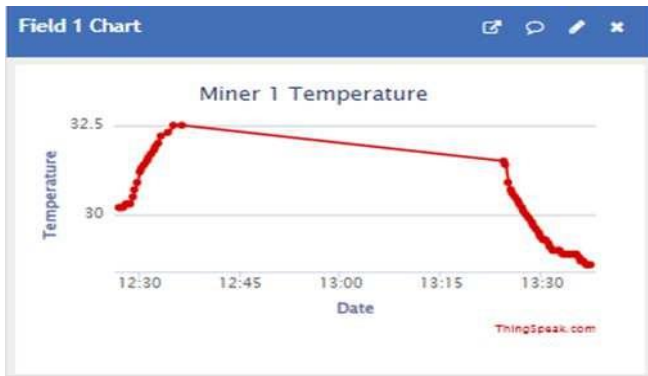


Fig 5(a): Mine temperature reading on ThingSpeak.



Fig 5(b): Mine humidity reading on ThingSpeak

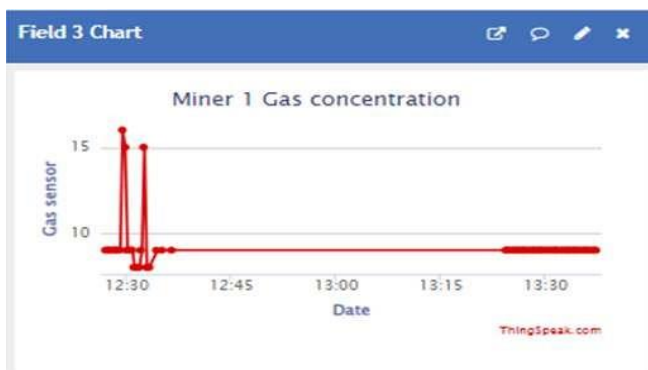


Fig 5(c): Mine gas reading on ThingSpeak

Fig.5(a) illustrates the change in mine temperature reading, Fig.5(b) and 5(c) shows the changes in humidity and gas concentration of miner 1 in ThingSpeak. After completing the verification process on ThingSpeak, users must authenticate their access by logging into the platform with their designated credentials. Fig.6 illustrates the login interface, where user authentication is enforced to ensure data security and system integrity. This authentication mechanism serves as a fundamental security measure to safeguard sensitive data and prevent unauthorized access. Upon successful login, users must validate network functionality by verifying the accurate transmission and real-time display of sensor readings within the platform.

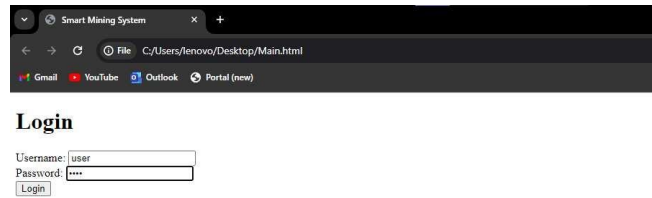


Fig 6: Login credential for the webpage

Miners Sensor Data



Miners Sensor Data

Temperature, Humidity, and Gas Level

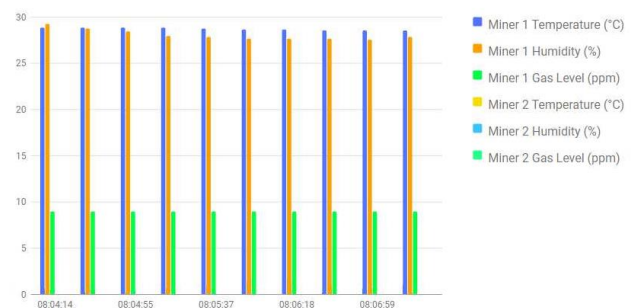


Fig 7: Readings shown on the webpage.

Fig.7 shows the miners reading web interface shows accurate information and the graph displayed reflects the fidelity of the values recorded by the sensors. The webpage displays all three parameters, which are the temperature, humidity as well as the gas levels, and it shows the value of each sensor with the help of graphical representation of each miner. The graph is color coded where each miner has an assorted color. On the graph's x-axis, the timing of the sensor readings is shown, and on the y-axis, the parameter's magnitude is shown. This tool is used for easy analysis and interpretation of data.

Handling network downtime in underground mining conditions is crucial for maintaining continuous safety monitoring. One effective strategy is implementing offline data storage within the helmet system itself. By integrating local storage solutions, such as onboard flash memory or SD cards within the NodeMCU microcontroller, sensor data can be temporarily stored when network connectivity is lost. Once the connection is restored, the stored data can be synchronized with the cloud platform (ThingSpeak) to ensure no critical information is lost. Additionally, using edge computing techniques, the helmet can process and analyze environmental data locally, triggering immediate safety alerts for miners even when real-time cloud access is unavailable. Implementing redundant communication channels, such as LoRaWAN or mesh networking, can further enhance data transmission reliability in deep underground environments where Wi-Fi signals may be weak or intermittent. These strategies ensure that miners

remain protected, and supervisors maintain access to essential safety data despite connectivity challenges.

The proposed smart helmet system demonstrates significant advancements over existing solutions by integrating multiple real-time monitoring features, enhanced communication protocols, and improved safety measures. Unlike traditional safety helmets that provide only physical protection, the proposed system incorporates IoT-enabled sensors for detecting hazardous gases (CO, CH₄, LPG), real-time localization using RF and Wi-Fi, and smart alarming mechanisms with LED indicators and buzzers. Compared to previous solutions that relied on standalone gas detection or limited connectivity, this system ensures seamless data transmission to both miners and control stations via cloud integration, enhancing situational awareness. Additionally, while existing smart helmets often face challenges in power efficiency and data privacy, the proposed design optimizes energy consumption through adaptive sensing mechanisms and employs secure data handling techniques. Furthermore, by leveraging Lora WAN and wireless sensor networks, the proposed system extends communication range in underground environments, addressing a major limitation of RF-based systems. Comparative analysis validates the effectiveness of the proposed helmet by demonstrating its capability to provide proactive hazard detection, minimize response time, and enhance overall miner safety through real-time alerts and continuous environmental monitoring.

V. CONCLUSION AND FUTURE DIRECTIONS

The proposed system provides a reliable solution for monitoring underground mining conditions using DHT11 and MQ135 sensors integrated with a NodeMCU (ESP8266). The collected data is transmitted wirelessly to the ThingSpeak cloud platform, enabling real-time monitoring through a miner readings webpage. This approach enhances safety protocols by continuously analyzing environmental parameters, detecting hazardous conditions, and allowing timely intervention. However, underground mining environments pose significant challenges, such as signal attenuation, interference, and network downtime, which can impact real-time data transmission and monitoring.

To further enhance the system, future work can explore hybrid communication models incorporating LoRa, Zigbee, or mesh networks to extend coverage and improve connectivity in underground tunnels. Additionally, the implementation of offline data storage with automated synchronization once connectivity is restored can ensure continuous data logging. The integration of AI-driven predictive analytics and machine learning algorithms can help in early hazard detection and proactive safety measures. Moreover, incorporating edge computing can reduce latency by processing critical data at the source, enhancing real-time decision-making. Future improvements can also focus on ruggedized sensor enclosures and power-efficient solutions to ensure long-term sustainability in harsh mining conditions.

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